

# I. CLASSIFIER MODEL

## Overview

The PinoyBot starts by accepting a single Filipino passage. It is then expected to tag each token in the passage with the appropriate language tag. The trained supervised machine learning model uses the extracted features to predict the language tag of every token and assigns the correct label. Once the model has been trained successfully, the predicted tags are returned as the output. These steps allow the bot to classify each token as Filipino (FIL), English (ENG), code-switched (CS), or other (OTH).

The annotated dataset consisted of manually labeled words completed by the group. To increase the amount of training data, annotated sentence IDs from Groups 110, 99, and 83 were also reviewed. The annotations were checked for consistency with the group's labeling guidelines, and some entries were corrected before being included in the final dataset.

Additional annotated sentence IDs reviewed

- Group 110 - Sentence IDs 2180 - 2200
- Group 99 - Initially shared Sentence IDs 1042 -1265 however, the correct Sentence IDs should be 1960-1980
- Group 83 - Sentence IDs 1640 - 1660

## Classifier

The group used a Random Forest classifier from the scikit-learn library to predict the language label of each token. This classifier was selected because it works well for classification tasks and can use different types of features to identify patterns in each token. By combining the predictions of multiple decision trees, it can make more accurate and reliable predictions than using a single decision tree. This makes it a suitable classifier for identifying tokens. The model was configured with 50 decision trees and a random state of 42 to produce consistent results during training.

## Features

- **Lexical Features**
  - The model utilizes these features to capture the identity and basic characteristics of each token. These include the original word, lowercase representation, and its length. These allow the model to learn association between frequently occurring tokens and ensure that all inputs are consistent.
- **Orthographic Features**
  - These describe how a word is written to the model. These include where a token starts with a capital letter, contains digits or is entirely digits, hyphens, or apostrophes. The model uses these to identify abbreviations, acronyms, expressions, and numerical tokens.
- **Morphological Features**
  - The model consists of prefixes, suffixes, and common Filipino affixes such as mag, nag, um, in, and ng. These are used to identify recurring word beginnings and endings that may indicate a particular classification.
- **Character Statistical Features**
  - This includes the number of vowels, uppercase letters, digits, and whether the token ends with a vowel. These features provide statistical information about the composition of a word, as different languages often have distinct characteristics and spelling patterns.

## Final Configuration

The group's PinoyBot uses a Random Forest classifier with 50 decision trees to classify each token. A random state of 42 was also used so that the data is split in the same way and the model produces consistent results every time it is trained. Before training, the dataset is cleaned by removing missing values and invalid labels. The data was split into 70% training, 15% validation, and 15% testing using stratified sampling to maintain the distribution of the language tags. The final model uses the lexical, orthographic, morphological, and character statistical features. These features were selected because they help the model identify different patterns in each token and improve its ability to classify them. This final configuration was chosen because it provides good classification performance while keeping the model simple and efficient.

## II. EVALUATION AND PERFORMANCE

After training the Random Forest classifier, its performance was evaluated using the validation and test sets. The evaluation was based on accuracy, precision, recall, and F1-score to determine how well the model identified each language label.

Table 1. Classification Performance (Test Set)

| Class | Precision | Recall | F1-score |
|-------|-----------|--------|----------|
| CS    | 1.00      | 0.17   | 0.29     |
| ENG   | 0.90      | 0.81   | 0.85     |
| FIL   | 0.97      | 0.99   | 0.98     |
| OTH   | 0.94      | 0.89   | 0.92     |

The model achieved high precision, recall, and F1-scores for FIL, ENG, and OTH, showing that it was able to identify these language labels accurately. In comparison, CS had the lowest performance. This may be because there were fewer CS samples in the dataset, and these words shared characteristics with both Filipino and English, making them more difficult for the model to identify correctly.

Table 2. Overall Accuracy

| Metric    | Validation | Test   |
|-----------|------------|--------|
| Accuracy  | 95.38%     | 95.85% |
| Precision | 95.26%     | 95.78% |
| Recall    | 95.38%     | 95.85% |
| F1-Score  | 95.20%     | 95.64% |

The overall evaluation scores were also consistent between the validation and test sets. The test set achieved 95.85% accuracy, 95.78% precision, 95.85% recall, and 95.64% F1-score, with only small differences from the validation results. This suggests that the model maintained similar performance when evaluated on unseen data.

Fig. 1. Dojo Results

CONFUSION MATRIX

|     | CS | ENG | FIL | OTH |
|-----|----|-----|-----|-----|
| CS  | 2  | 0   | 0   | 0   |
| ENG | 0  | 60  | 9   | 4   |
| FIL | 0  | 1   | 429 | 12  |
| OTH | 0  | 3   | 12  | 70  |

OVERALL ACCURACY  
93.19%

WEIGHTED AVG. F1  
0.9314

CLASS METRICS

| Class | Recall | Precision | F1 Score | Support |
|-------|--------|-----------|----------|---------|
| CS    | 1.0000 | 1.0000    | 1.0000   | 2       |
| ENG   | 0.8219 | 0.9375    | 0.8759   | 73      |
| FIL   | 0.9706 | 0.9533    | 0.9619   | 442     |
| OTH   | 0.8235 | 0.8140    | 0.8187   | 85      |

Our bot achieved a 93.19% accuracy and a 0.9314 weighted F1-score on the Dojo test cases. Most predicted labels matched expected labels, correctly identifying 429 out of 442 FIL, 60 out of 72 ENG, 2 out of 2 CS, and 70 out of 85 OTH tokens respectively.

### III. CHALLENGES

Annotating almost 42,000 words with only four group members took a lot of time. Some words were also hard to label as FIL, ENG, or OTH because they could fit more than one category. This was especially true for *jejemon* typings, since it was sometimes unclear whether they should still be labeled as FIL or placed under OTH. When coordinating with other groups, some of the annotations were different from how our group understood the labeling guidelines. We reviewed and corrected some entries, and in some cases decided not to use annotations that were not consistent with our labeling approach. Because of the large number of words, keeping the labels consistent throughout the dataset was also a challenge.

Another challenge was deciding on the machine learning algorithm to use for the bot. The group had to test different classifiers and features before agreeing on the combination that gave the best results. The bot also had difficulty identifying CS words because they shared characteristics with both Filipino and English.

Starting the annotation process earlier would give the group more time to complete and review the dataset. It would also allow enough time to compare and exchange annotations with other groups. Regular discussions within the group can also ensure that everyone follows the same labeling guidelines. In addition, trying different features and classifiers may further improve the bot's ability to identify CS and borrowed words.

#### IV. TABLE OF CONTRIBUTIONS

| Member                        | Contributions                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------|
| Catapang,<br>Nathan James B.  | Model Development, Testing, Report Paper, and Data Annotation                     |
| Hernaez,<br>Raeka Estrelle R. | Dataset Preparation and Preprocessing, Testing, Report Paper, and Data Annotation |
| Serrano,<br>Paul Rhazzel      | Feature Engineering, Testing, Report Paper, and Data Annotation                   |
| Tamayo,<br>Chelsea Nichole C. | Implementation and Integration, Testing, Report Paper, and Data Annotation        |

#### V. REFERENCES

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